+1 (585) 303-4037 Mikhail Saleev mikhailsaleev123@gmail.com

Education

University of Notre Dame | B.A. Computer Science and Economics | Concentration in Financial Economics & Econometrics | '23 – '27 | 3.83 GPA | *Dean's List:* Fall '23, Spring '24

Coursework: Machine Learning, Intro to AI, Design & Analysis of Algorithms, Probability, Multivariable Calculus, Linear Algebra & Differential Equations, Partial Differential Equations, Data Structures, Theory of Computing, Econometrics

Study Abroad: International Economics Berlin, '24; Rome Summer Program, '25

Publications

- **Deep Learning Estimation of the Number and Locations of Surfaces for Underwater ISAC**. *Submitted ICASSP'27*
- **Ultra-Wideband Indoor Localization and Bistatic Deep Learning-Based Mapping for ISAC**. *Submitted ICC'27*

Research Experience

Arikan Lab | *Undergraduate Research Assistant* | Aug. '25 – Present

- Extended U-COTANS to an end-to-end, generalizable ISAC method for transformer-based reflective boundary estimation in underwater acoustic environments with up to 4 boundaries, reaching state-of-the-art accuracy across 6-28 dB SNR
- Designed an amplitude-weighted COTANS framework for localizing boundaries from multiple competing arrivals
- Created a geometry-based echo labeling algorithm mapping neural net boundary estimates to TOAs across 5 receivers

Zhang Lab | *Undergraduate Research Assistant* | May '26 – Present

- Identified decoder-flow co-adaptation in LatentRxnFlow that caused the decoder to recover ~52% of USPTO-50K test products without the flow network
- Working to align the chemical latent space geometry with the pairwise flow matching objective to enable robust prediction of reaction-specific atom trajectories by the velocity network
- Using physics-based rewards from AIMNet2 to generate plausible intermediate reaction states for GRPO fine-tuning

Drone Response Startup | *Undergraduate Research Assistant* | Jan. '25 – Aug. '25

- Annotated a dataset of 200+ images across 7 semantic classes via CVAT.ai to fine-tune the drone image segmentation model, leading to 8% improvement in the model's prediction quality
- Reduced technical debt by refactoring legacy Numpy, Scikit, Keras, and Tensorflow code and writing documentation

Additional Experience

University of Notre Dame | *Teaching Assistant & Learning Resource Center Tutor* | Aug. '24 – May '26

- 1 of 3 Discrete Math TAs for Fall '24. Wrote and graded problem sets, holding weekly office hours for a class of 51 students
- Tutored 50+ students in Calculus and Linear Algebra, driving a 10% improvement in exam scores over prior semester

Visa & University of Notre Dame | *Visa Fintech Foundations Program Scholar* | Oct. '24 – Jan. '25

- Led a team of 6 to pitch a proposal in favor of Wholesale token-based DLT CBDCs to Visa's senior management, placing 3rd in the corporate case competition. Attended a 6-week intensive fintech lecture series

Projects

- Built and deployed TradeSphere.ai, a full-stack web app for implementing trading strategies, analyzing portfolio performance with yfinance data backbone, and providing AI-driven insights. Ran 1000+ portfolio simulations
- Built an agentic RAG pipeline processing research drafts via schema-constrained Qwen3:32B calls to Nebius' Endpoint running Dockerized Ollama. Implemented a local FAISS search over SPECTER2 arXiv abstract embeddings to suggest architectural and conceptual improvements based on related research work
- Created an NFA-based regular expression matcher supporting operators and backreferences to emulate Unix grep/sed

Skills

- *Programming Languages*: Python, Java, C, C#, JavaScript, TypeScript, SQL
- *Technologies*: PyTorch, TensorFlow, NumPy, Pandas, SciPy, Flask, Next.js, Tailwind, Bootstrap, Git, AWS Bedrock
- *Other*: Poker, Piano, Chess, Ping Pong, Languages (*English, Russian, French, Latin*)